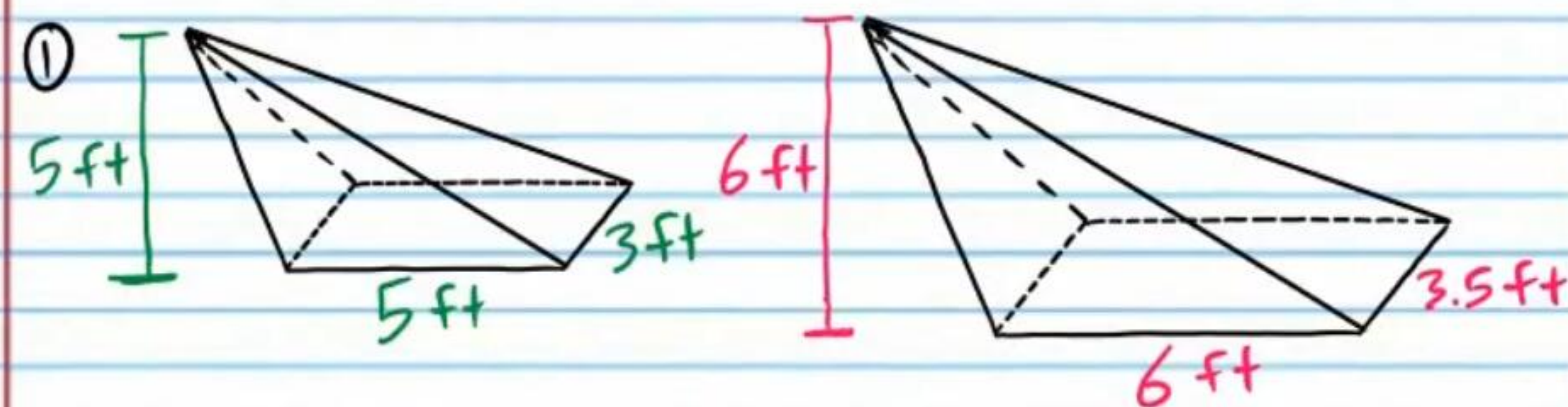


Practice Geometry Test VI (Version 2)

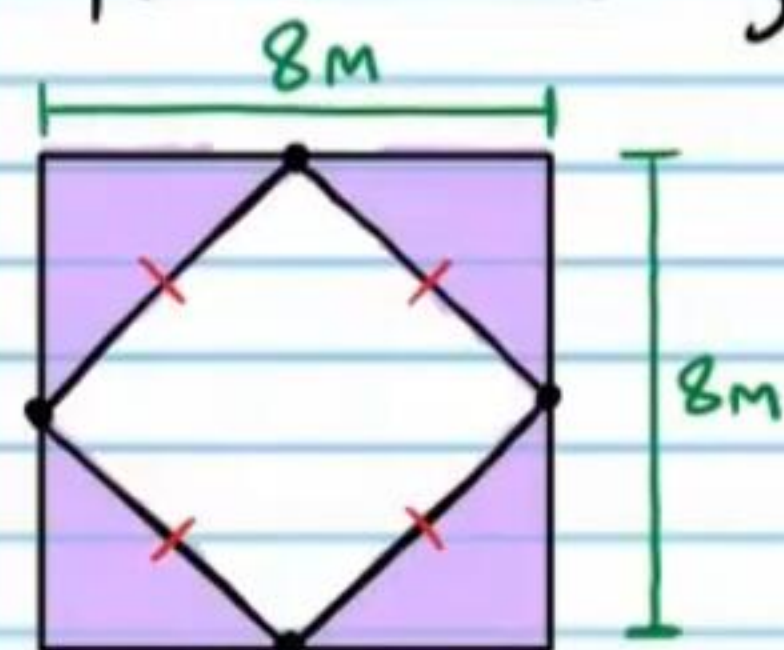
Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- V) Each problem is worth four points for a total of 100 points.

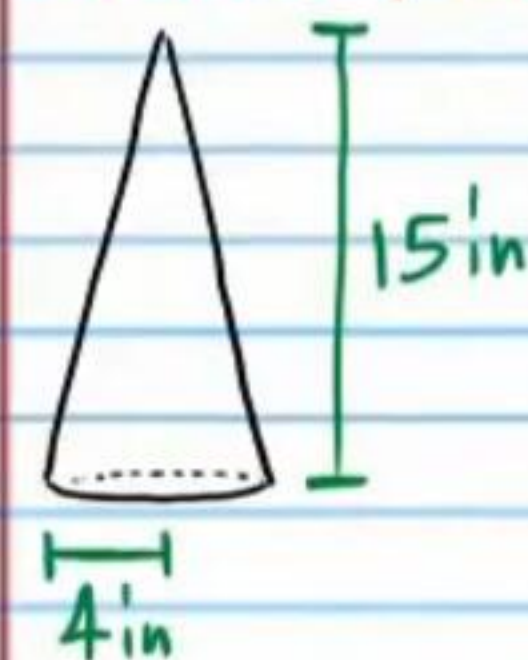
Determine if the pair of figures below contains similar solids. Write the ratios that prove your answers.



- ② Find the probability that a randomly chosen point in the figure will lie in the shaded region.

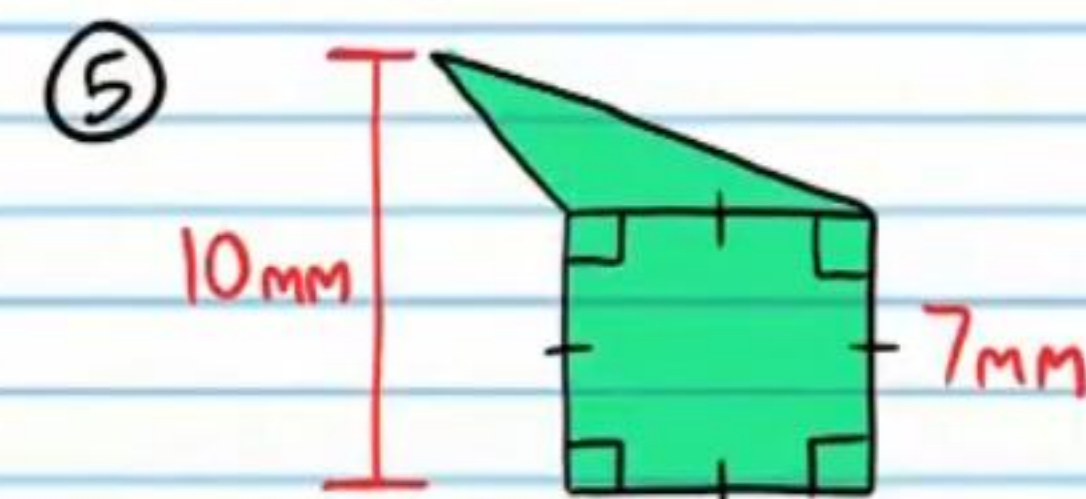
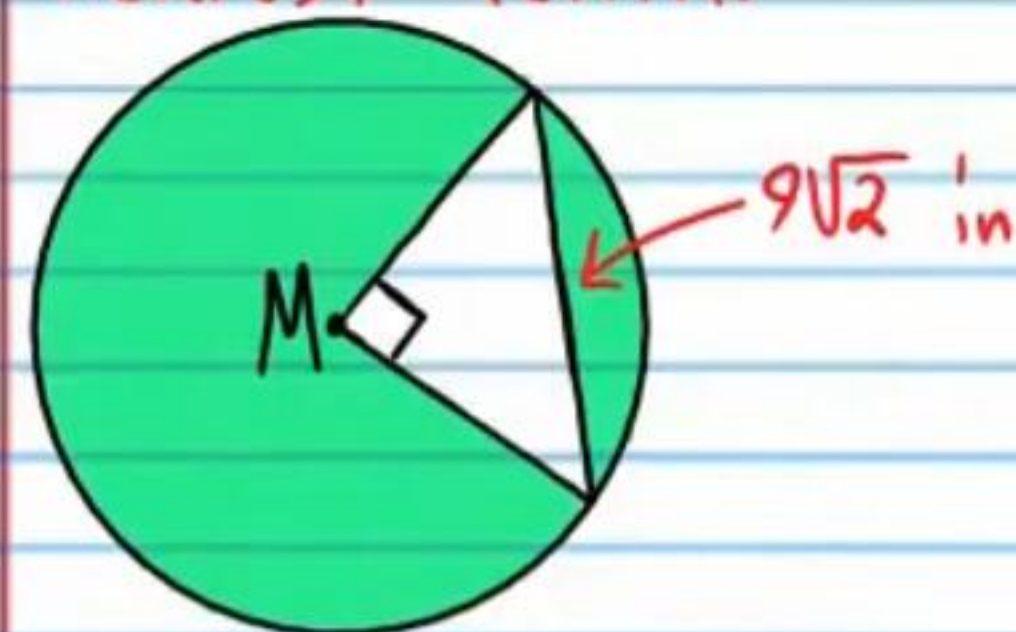


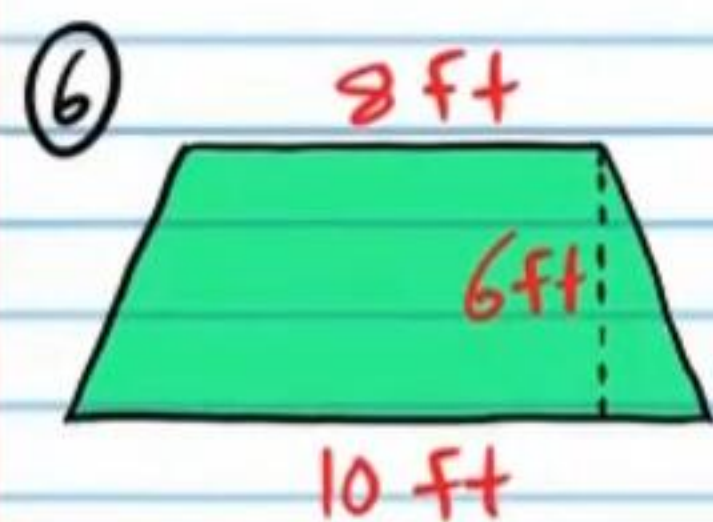
- ③ Find the surface area of the right cone below. Round your answer to the nearest tenth.



Find the area of each shaded region below.

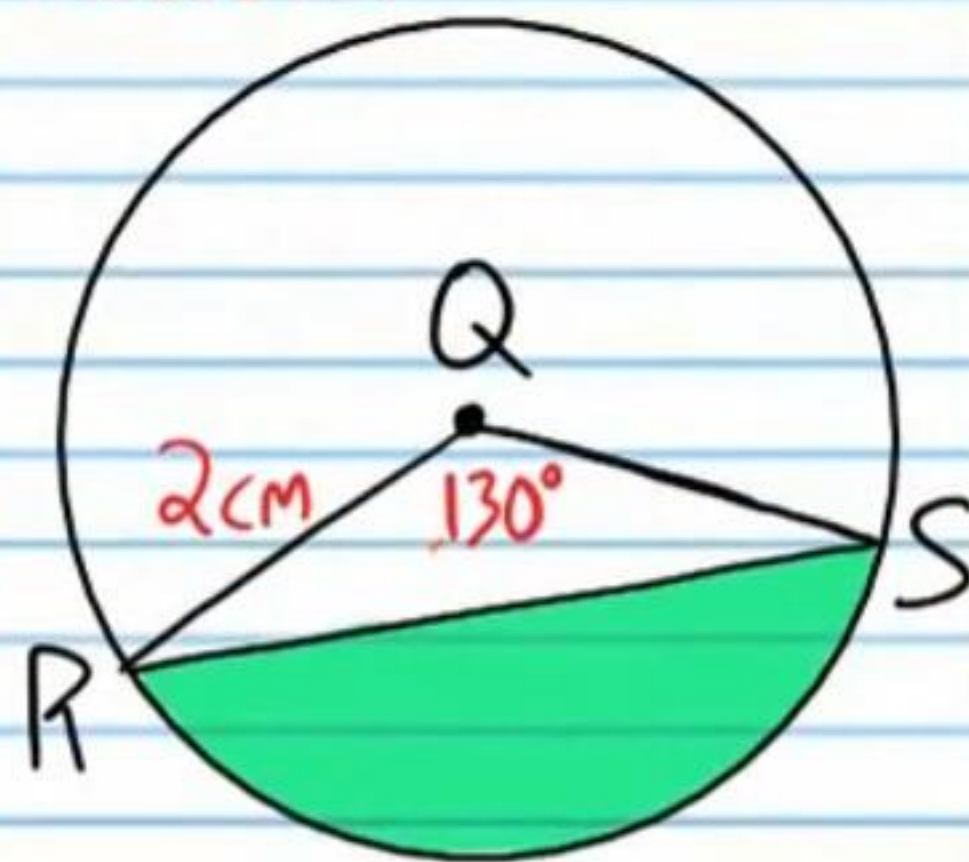
- ④ M is the center. Round your answer to the nearest tenth.





Use ⓐ below to answer problem #'s 7-9.

⑦ Find the area of sector RQS . Round your answer to the nearest ten thousandth.

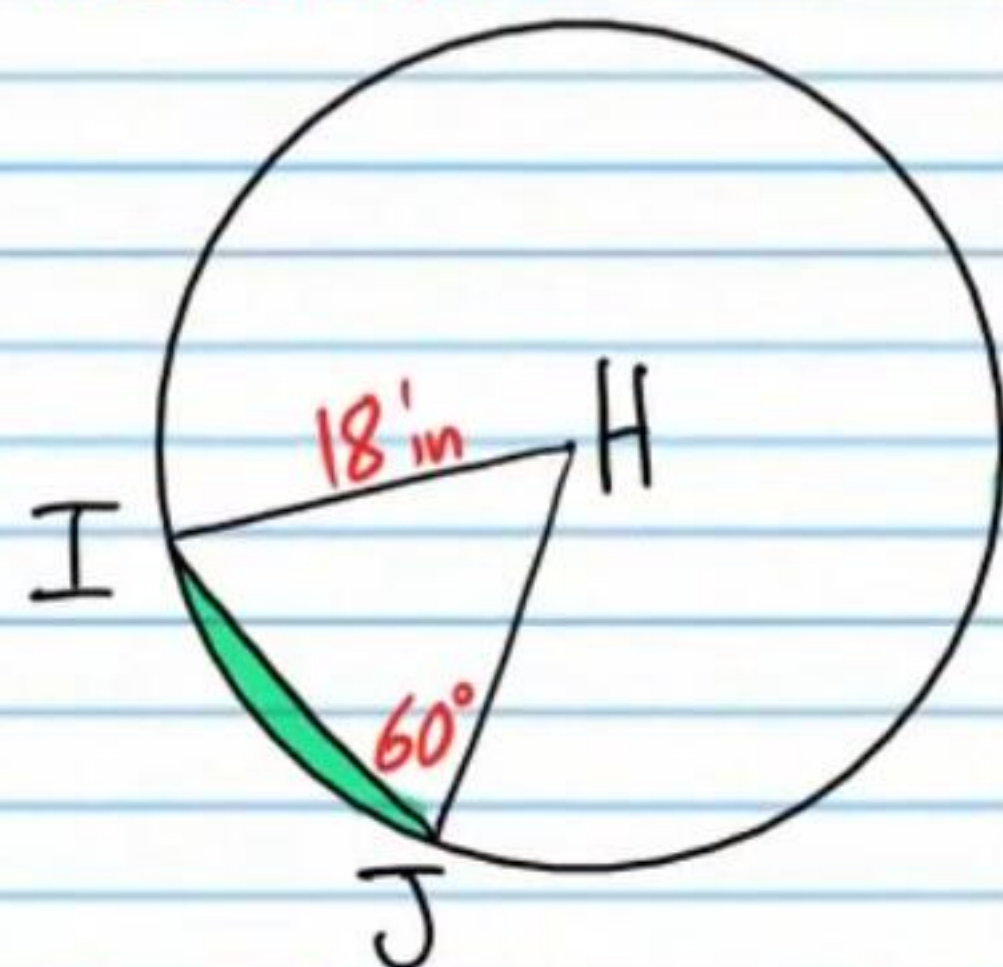


⑧ Find the area of $\triangle QRS$. Round your answer to the nearest ten thousandth.

⑨ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

Use Ⓐ below to answer problem #'s 10-12.

- ⑩ Find the area of sector IHJ. Round your answer to the nearest ten thousandth.



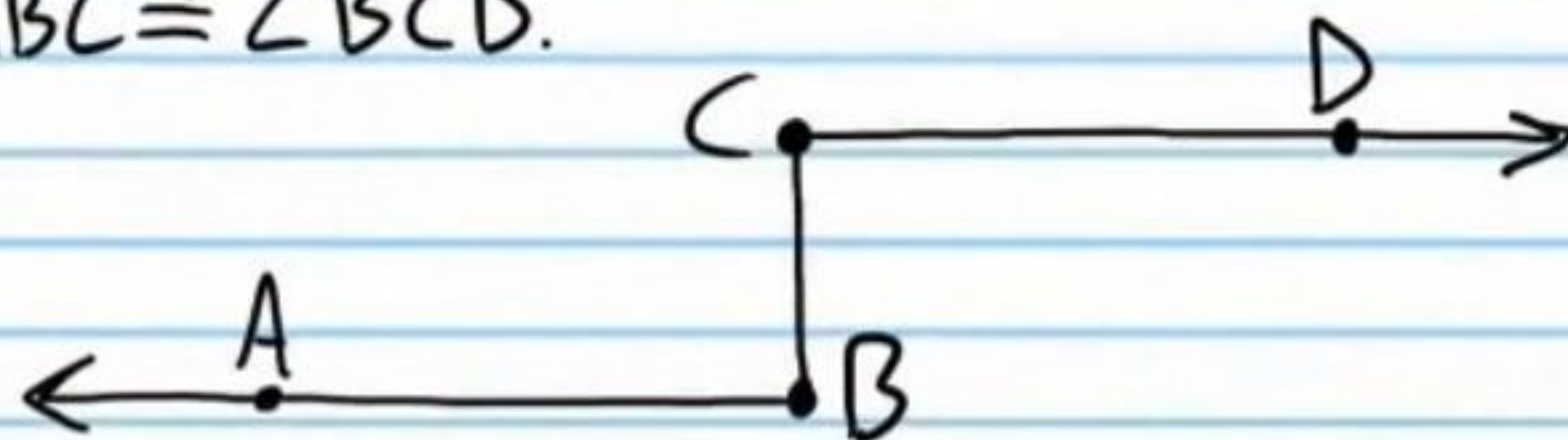
- ⑪ Find the area of $\triangle IHJ$. Round your answer to the nearest ten thousandth or write an exact expression.

- ⑫ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

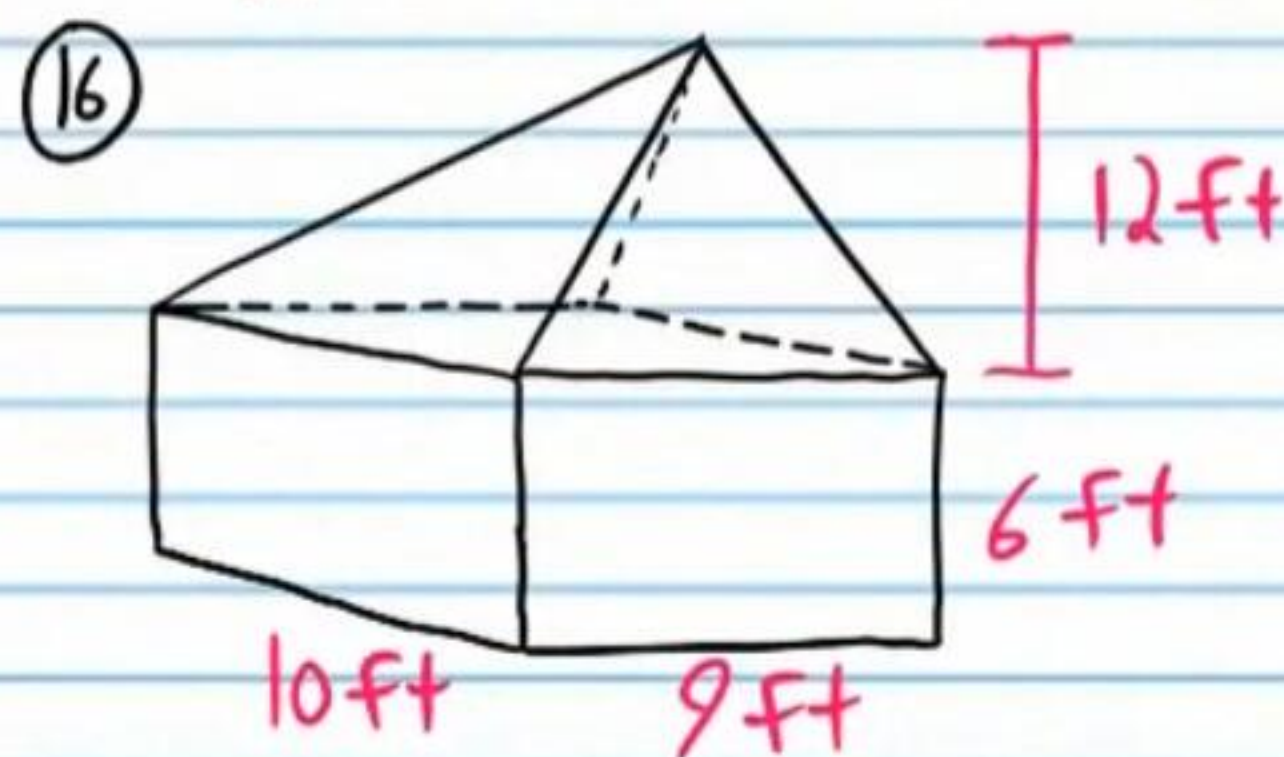
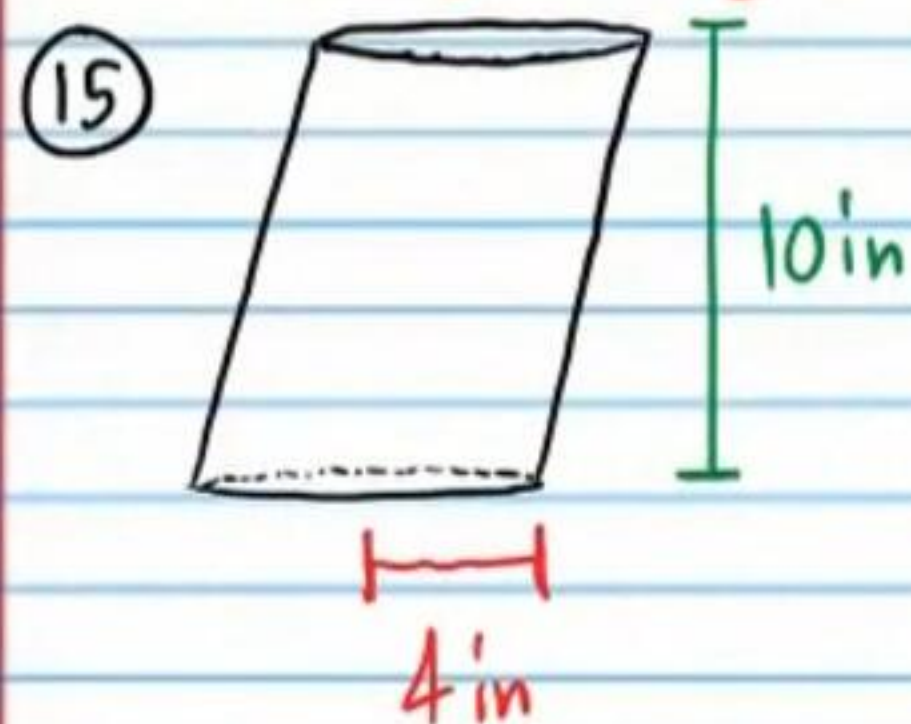
- ⑬ The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the Volume of the larger solid, find the volume of the smaller solid.

$$SF = \frac{1}{2} \quad V = 9 \text{ in}^3$$

- ⑭ Write a flowchart proof for the following statement: If $\overrightarrow{BA} \perp \overrightarrow{BC}$ below and $\overrightarrow{BC} \perp \overrightarrow{CD}$, then $\angle ABC \cong \angle BCD$.

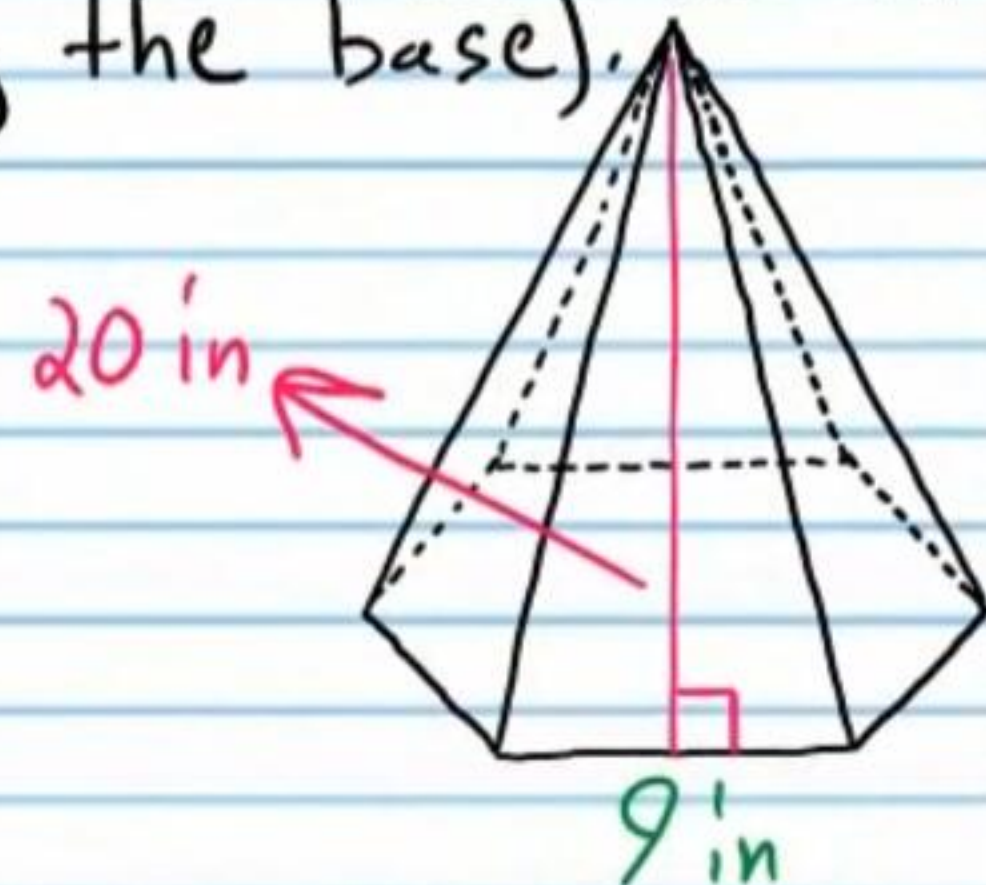


Find the volume of the following figures. Round your answers to the nearest tenth if necessary. Assume right angles where apparent.



Use the regular pyramid below to answer problem #'s 17-19.

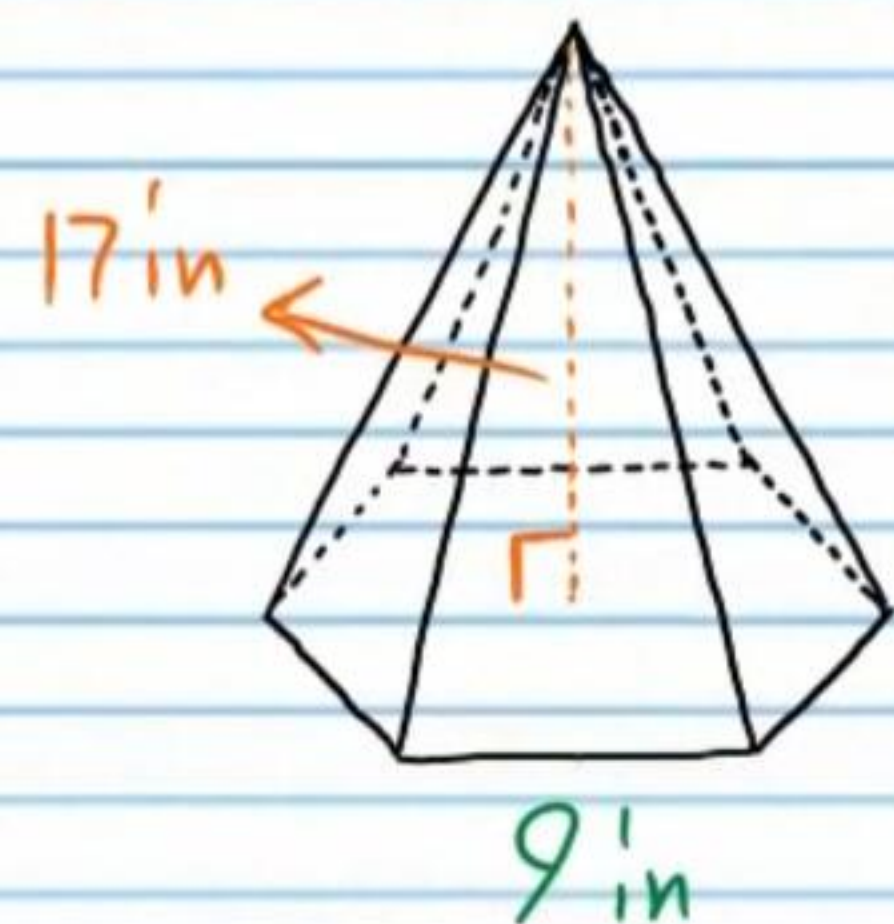
⑰ Find the total area of all of the sides of the pyramid (not including the base).



⑱ Find the area of the base of the pyramid. Round your answer to the nearest ten thousandth.

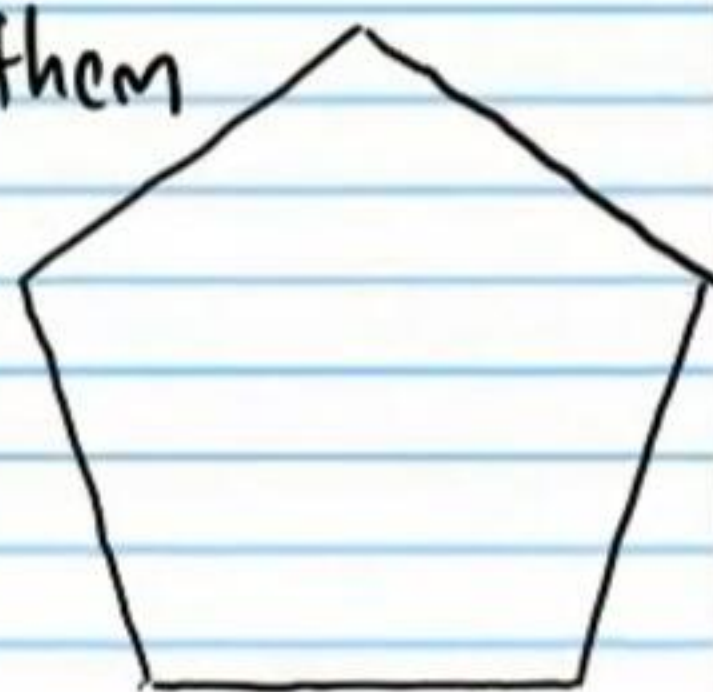
⑲ What is the total surface area of the pyramid? Round your answer to the nearest tenth.

⑳ Find the volume of the pyramid in problem #'s 17-19 if its height is 17 inches. Round to the nearest tenth.



Use the regular polygon below to answer problem #'s 21-22.

21) Find the length of the i) apothem and the ii) side lengths if the radius is 5mm. Round to the nearest ten thousandth.



22) Find the area of the polygon. Round your answer to the nearest tenth.

Use the two-column proof below to do problem # 23.

If B and C lie between A and D (collinear) as shown on the segment below, then $AD = AB + BC + CD$.

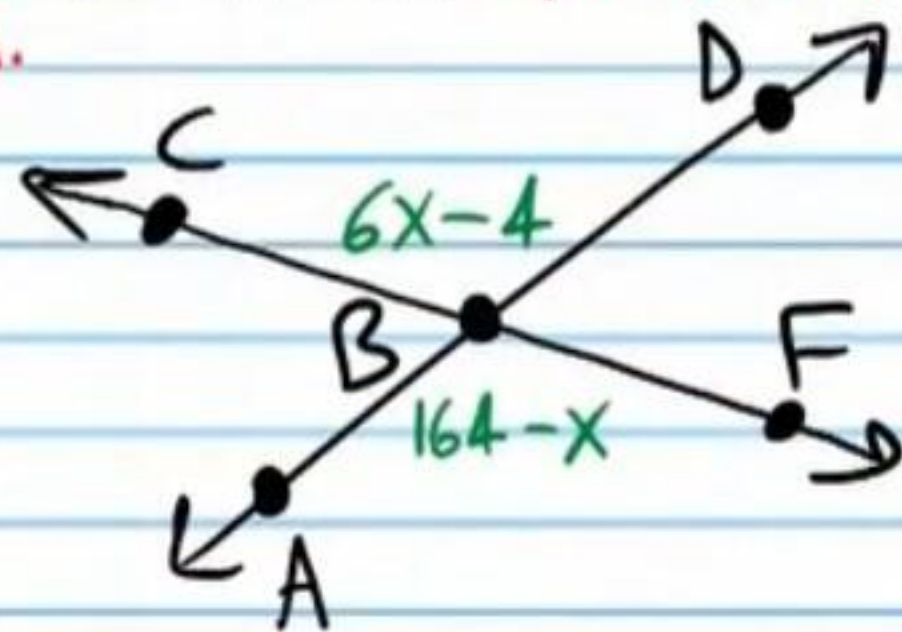


Statement	Reason
I) B and C lie between A and D.	Given.
II) $AC = AB + BC$	Seg. Add. Post.
III) $AD = AC + CD$	Seg. Add. Post.
IV) $AD = AB + BC + CD$	Subs. Prop of =.

23) Convert the two-column proof above to paragraph form.

Use the statement below to do problem #'s 24-25.

If $\angle CBD$ and $\angle ABF$ are vertical angles in the figure below, then $x=24$. ~~★~~ You must use the Vertical Angles Theorem.

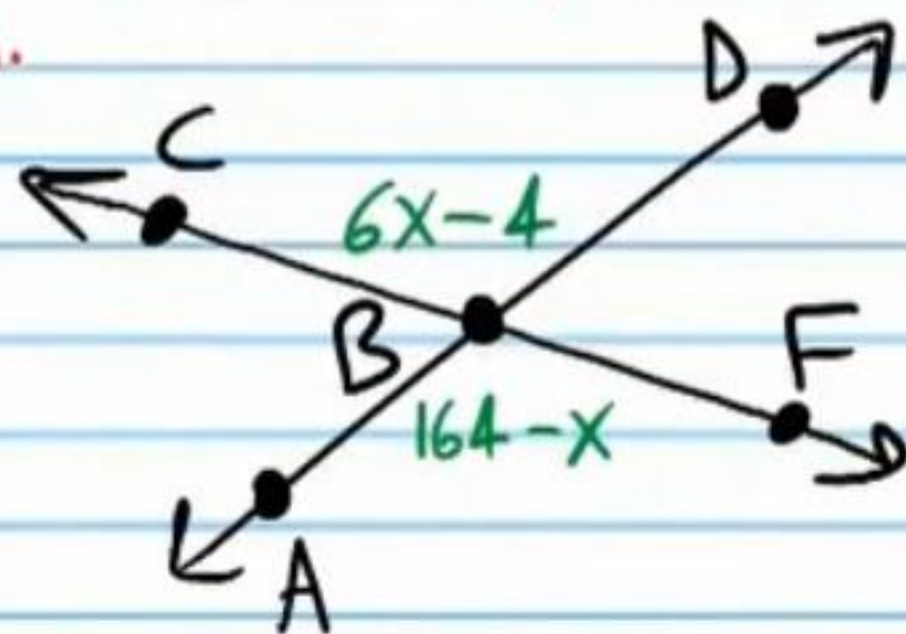


②④ Write a flowchart proof for the statement above.

②⑤ Write a paragraph proof for the statement.

Use the statement below to do problem #'s 24-25.

If $\angle CBD$ and $\angle ABF$ are vertical angles in the figure below, then $x=24$. ~~★~~ You must use the Vertical Angles Theorem.



②④ Write a flowchart proof for the statement above.

②⑤ Write a paragraph proof for the statement.

Practice Geometry Test VI (Version 2) Answers

① $\frac{6}{5} = \frac{6}{5} \neq \frac{3.5}{3}$
 $1.2 = 1.2 \neq 1.1\bar{6}$
 Not similar

② $\frac{1}{2}$

③ 245.3 in^2

④ 214 in^2

⑤ 59.5 mm^2

⑥ 54 ft^2

⑦ 4.5379 cm^2

⑧ 1.5321 cm^2

⑨ 3 cm^2

⑩ 169.646 in^2

⑪ 140.2961 in^2 or $81\sqrt{3} \text{ in}^2$

⑫ 29.3 in^2

⑬ $\frac{9}{8} \text{ in}^3$ or $1\frac{1}{8} \text{ in}^3$ or 1.125 in^3

⑭ $\vec{BA} \perp \vec{BC}$

Given.



$\angle ABC$ is a right angle.

Def. of $\perp$.

$\vec{BC} \perp \vec{CD}$

Given.



$\angle BCD$ is a right angle.

Def. of $\perp$.

$\angle ABC \cong \angle BCD$
 Rt $\angle \cong$ Thm.

⑮ $160\pi \text{ in}^3$ or 502.7 in^3

⑯ 900 ft^3

⑰ 540 in^2

⑱ 210.4442 in^2

⑲ 750.4 in^2

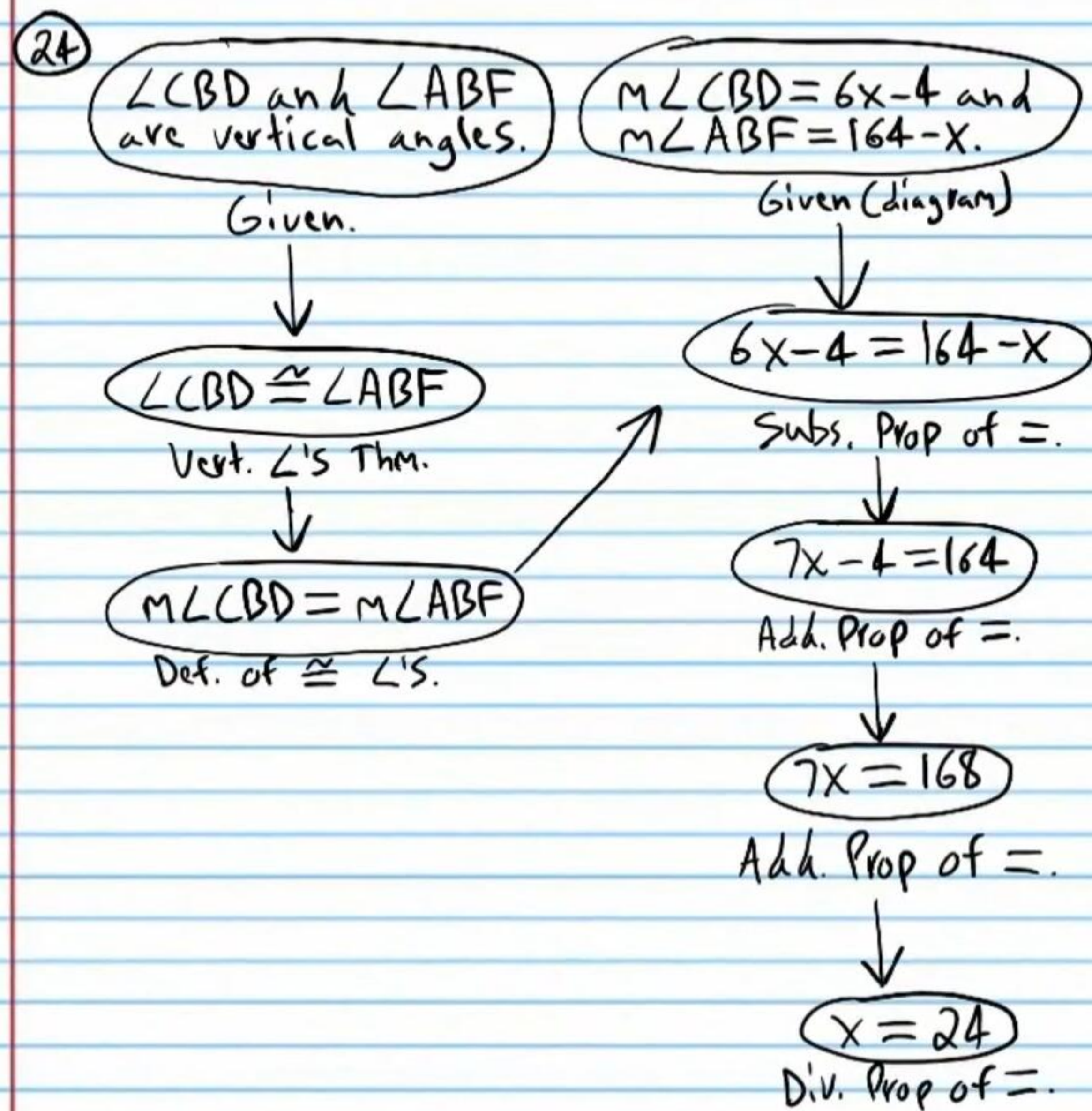
⑳ $1,192.5 \text{ in}^3$

㉑ i) Apothem: 4.0451 mm

ii) Side Length: 5.8779 mm

㉒ 59.4 mm^2

②③ It is given that B and C lie between A and D. By the Segment Addition Postulate, $AC = AB + BC$ and $AD = AC + CD$. Using the Substitution Property of Equality, we can conclude that $AD = AB + BC + CD$.



②⑤ It is given that $\angle CBD$ and $\angle ABF$ are vertical angles. It is given in the diagram that $m\angle CBD = 6x - 4$ and $m\angle ABF = 164 - x$. By the Vertical Angles Theorem, $\angle CBD \cong \angle ABF$ and by the definition of congruent angles $m\angle CBD = m\angle ABF$. Using this equation along with the Substitution Property of Equality, we can write the equation $6x - 4 = 164 - x$. Using the Addition Property of Equality, we can say that $7x - 4 = 164$ and $7x = 168$. Finally, by the Division Property of Equality, $x = 24$.